PES2UG24CS118 BHUVI DANGE SECTION B

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-01	Functional	The system shall allow a resident to register an account with name, unit/apartment number, and contact details.	High	Given valid inputs, submitting the registration form creates an account and displays a confirmation within 2 seconds.	Registration is the entry point for all other functionality.
FR-02	Functional	The system shall allow a registered resident to view equipment details and current availability status.	High	Selecting an equipment item displays name, category, condition, and an "Available"/"Unavailable" status that matches the backend record.	Residents need visibility before requesting a borrow.
FR-03	Functional	The system shall allow a resident to submit a borrowing request only if the equipment is available and the resident has no overdue equipment.	High	A request is accepted only when both checks pass; if either fails, the system blocks submission and shows the specific reason (item unavailable / resident has overdue item).	Enforces the core eligibility rule stated in the scenario.

FR-04	Functional	The system shall allow the administrator to approve a pending borrowing request and record the issue of equipment to the resident.	High	On approval, request status changes to "Issued," an issue date is logged, and the equipment's availability flag is set to false.	Admin approval is the gate before physical handover.
FR-05	Functional	The system shall allow the administrator to record equipment return, log its condition, and charge a repair fee if it is returned damaged; it shall also allow a resident to reserve unavailable equipment and receive a notification when it becomes available.	Medium	Recording a return sets status to "Returned" and condition (Good/Damaged); if "Damaged" is selected, a repair fee field becomes mandatory. A reservation on unavailable equipment triggers a notification to the reserving resident within 5 minutes of the item's status changing to available.	Covers end-of-cycle handling (return/damage) and the reservation-notification flow explicitly described in the scenario.
NFR-01	Nonfunctional	The system shall send a return-due-date reminder to the resident no later than 24 hours before the due date, with at least 99% delivery success.	High	A delivery log shows the reminder timestamp is ≥24h before due date for 99% of sampled borrow records.	Prevents overdue equipment and supports the eligibility check in FR-03.

NFR-02	Nonfunctional	The system shall restrict approval of borrowing requests, issue/return recording, and repair-fee charges to authenticated administrator accounts only.	High	Attempting any of these three actions from a non-administrator session returns an authorization error and the action is not logged as completed.	Protects data integrity and prevents residents from self-approving requests.

2. Actors & Use Cases (for the UML diagram)

Actors :

- **Resident** registers, browses, borrows, reserves, receives notifications
- **Library Administrator** approves requests, issues/returns equipment, charges fees
- **Notification System** external system that dispatches reminders/availability alerts

Use Cases :

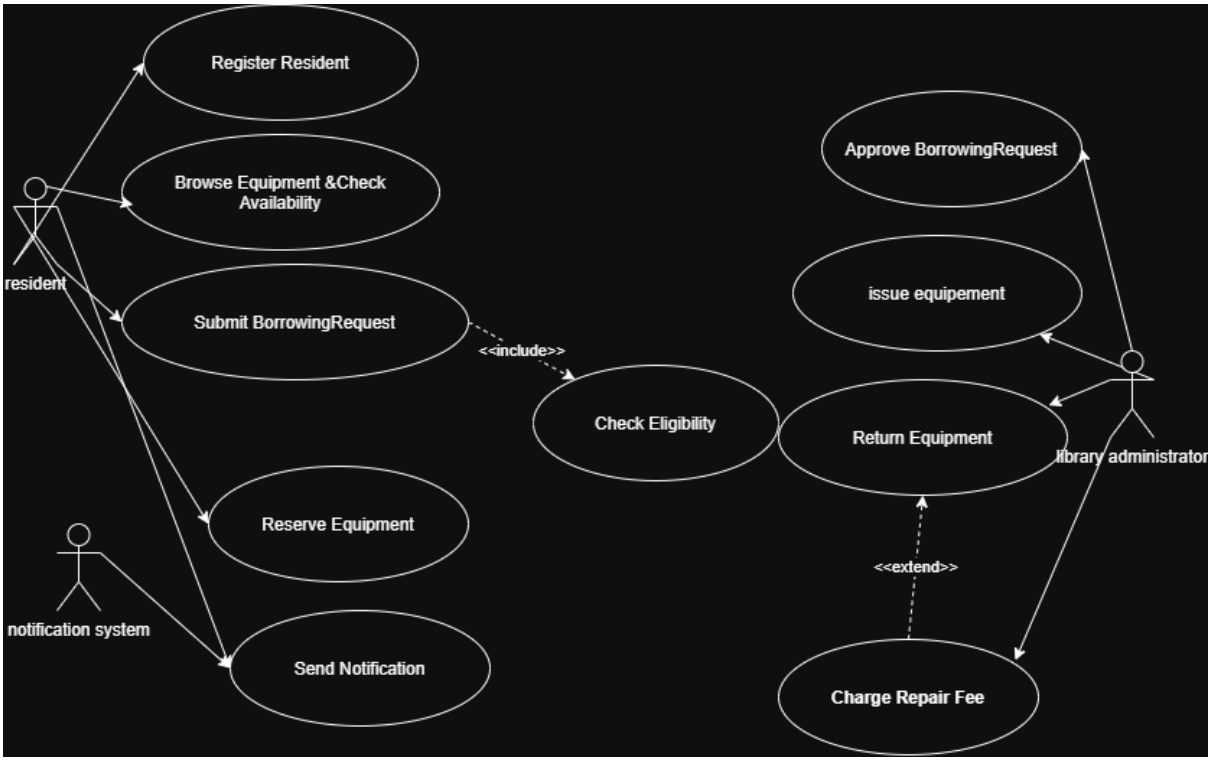
ID	Use Case	Actors
UC-01	Register Resident	Resident
UC-02	Browse Equipment & Check Availability	Resident
UC-03	Submit Borrowing Request	Resident
UC-04	Check Eligibility (availability + no overdue)	(included)
UC-05	Approve Borrowing Request	Administrator
UC-06	Issue Equipment	Administrator
UC-07	Return Equipment	Administrator
UC-08	Charge Repair Fee	Administrator
UC-09	Reserve Equipment	Resident
UC-10	Send Notification (reminder / availability alert)	Notification System, Resident

Relationships:

- UC-03 «include» UC-04 -every borrowing request always triggers the eligibility check.

- UC-08 «extend» UC-07 -Charge Repair Fee only fires conditionally, when the returned item is marked damaged.

Associations: Resident<->(UC-01, UC-02, UC-03, UC-09, UC-10); Administrator←→(UC-05, UC-06, UC-07, UC-08); Notification System<->(UC-10).



3. Use-Case Flow Document

Use Case: Submit & Process Borrowing Request

Main Success Scenario:

1. Resident selects an equipment item and chooses "Request to Borrow."
2. System checks equipment availability and the resident's overdue status (includes UC-04: Check Eligibility).
3. Both checks pass; system creates a borrowing request with status "Pending" and notifies the administrator.
4. Administrator reviews the request and approves it.
5. System updates request status to "Issued," logs the issue date, and sets equipment availability to false.
6. System sends the resident a confirmation and schedules a return-date reminder.
7. On the return date, system sends a reminder notification to the resident (24 hours prior).
8. Resident returns the equipment to the administrator.
9. Administrator records the return and inspects the equipment's condition.
10. Use case ends successfully; equipment availability is set to true.

Alternate Flow 2a. Eligibility Check Fails

- 2a1. System displays the specific reason: "Equipment unavailable" or "You have overdue equipment."
- 2a2. Resident is blocked from submitting until the condition is resolved (e.g., prompted to reserve the item instead via UC-09).

9a. Equipment Returned Damaged

- 9a1. Administrator marks condition as "Damaged."
- 9a2. System requires the administrator to enter a repair fee amount.
- 9a3. System charges the fee to the resident's account and notifies them of the charge and reason.